May 7

- Zoom office hrs. next week
  - 1-3 pm Monday
  - 9-10 am Tuesday

- PSP due Sunday 11:59 pm, submit through email and cc group members
  - Will be posted to Gradescope
  - Graded similarly to HW

- final is cumulative, mostly on rings

- Thm. 15.1.4 If $\phi: R \to S$ is a ring homomorphism and $B \subset S$ is an ideal, then $\phi^{-1}(B) = \{r \in R : \phi(r) \in B\}$ is an ideal of R

  Proof

  We apply the ideal test.

  Nonempty: Since B is an ideal, $0_S \in B$. Since $\phi: (R,+) \to (S,+)$ is a group homomorphism, $\phi(0_R) = 0_S$. Thus $0_R \in \phi^{-1}(B)$, so $\phi^{-1}(B) \neq \emptyset$.

  Closure under subtraction: Let $x, y \in \phi^{-1}(B)$ and consider $\phi(x-y) = \phi(x) - \phi(y)$. Since $\phi(x), \phi(y) \in B$ and B is an ideal, $\phi(x) - \phi(y) \in B$. Thus $\phi(x-y) \in B$. $\phi(x+(-y)) = \phi(x) + \phi(-y) = \phi(x) - \phi(y)$

  Multiplicative Absorption: Let $r \in R$ and $x \in \phi^{-1}(B)$. We consider rx.
  We want $rx \in \phi^{-1}(B)$. Thus we want $\phi(rx) \in B$. Now since ϕ respects mult. $\phi(rx) = \phi(r) \cdot \phi(x)$. Since $\phi(x) \in B$ and B has mult. absorption, $\phi(r)\phi(x) \in B$. Then $rx \in \phi^{-1}(B)$. $\square$

- Cor. 15.2 $\ker \phi$ is an ideal

  Proof

  Note $\ker \phi = \{r \in R : \phi(r) = 0_S\} = \phi^{-1}(0)$, and $\{0_S\}$ is an ideal of S.

- Thm. 15.1.2 $\phi(R)$ is a subring of S but not generally an ideal

  $\phi(R) = \{s \in S : s = \phi(r) \text{ for some } r \in R\}$

Exercise Prove that if F is a field and $I \subset F$ is an ideal, then $I = \{0\}$ or $I = F$.

  Proof

  We apply the subring test.

  Nonempty: $0_R \in R$ and $\phi(0_R) = 0_S$, so $\phi(R) \neq \emptyset$.

  Closure under subtraction: Let $u, v \in \phi(R)$ and consider $u - v$. Now $u = \phi(a)$ and $v = \phi(b)$ for some $a, b \in R$. Thus $u - v = \phi(a) - \phi(b) = \phi(a-b)$. Since $a - b \in R$, $\phi(a-b) = u - v \in R$.

Closure under multiplication: Consider $u \cdot v = \phi(a)\phi(b) = \phi(ab)$, since ϕ is a ring homomorphism. Thus $uv \in \phi(R)$.

We conclude $\phi(R)$ is a subring of S. $\square$

Ex. $\iota : \mathbb{Z} \hookrightarrow \mathbb{Q}$ given by $\overset{\text{in } \mathbb{Z}}{a} \hookrightarrow \overset{\text{in } \mathbb{Q}}{a}$

Thus $\mathbb{Z} = \iota(\mathbb{Z})$ is a subring of $\mathbb{Q}$. Note: it is not an ideal

- F.I.T if $\phi : R \to S$ is a ring homomorphism, then $R/\ker\phi \cong \phi(R)$ with the isomorphism given by $r + \ker\phi \xmapsto{\bar{\Phi}} \phi(r)$.

Proof

We have already seen that $R/\ker\phi \cong \phi(R)$ as additive groups. So we just need to show that $\bar{\Phi}$ respects mult. Let $r + \ker\phi, s + \ker\phi \in R/\ker\phi$. Then $\bar{\Phi}((r+\ker\phi)(s+\ker\phi)) = \bar{\Phi}(rs + \ker\phi) = \phi(rs)$.

Now $\phi(rs) = \phi(r)\phi(s) = \bar{\Phi}(r+\ker\phi) \cdot \bar{\Phi}(s+\ker\phi)$. Thus $\bar{\Phi}((r+\ker\phi)(s+\ker\phi)) = \bar{\Phi}(r+\ker\phi)\bar{\Phi}(s+\ker\phi)$. $\square$

Ex. $\mathbb{Z}[\varepsilon] = \{a + b\varepsilon : a, b \in \mathbb{Z}, \varepsilon^2 = 0\}$ implicit: $\varepsilon \neq 0$

Define $\varphi : \mathbb{Z}[x] \longrightarrow \mathbb{Z}[\varepsilon]$ by $\varphi(x) = \varepsilon$. I.e $\varphi(f(x)) = f(\varepsilon)$.

$\varphi(\mathbb{Z}[x]) = \mathbb{Z}[\varepsilon]$, so φ is surjective. One can show $\ker\varphi = \langle x^2 \rangle$.

$a + bx \longrightarrow a + b\varepsilon$

F.I.T says $\mathbb{Z}[x]/\langle x^2 \rangle \cong \mathbb{Z}[\varepsilon]$. Since $x \cdot x \in \langle x^2 \rangle$ but $x \notin \langle x^2 \rangle$, we see $\langle x^2 \rangle$ is not prime. Notice $\varepsilon \cdot \varepsilon = 0$, so we have zero-divisors. In $\mathbb{Z}[x]/\langle x^2 \rangle$ we have $x \cdot x = 0$.

Ex. $\mathbb{Z}[i]$

Define $\psi : \mathbb{Z}[x] \longrightarrow \mathbb{Z}[i]$ by $\psi(x) = i$, I.e $\psi(f(x)) = f(i)$

$\psi(\mathbb{Z}[x]) = \mathbb{Z}[i]$, $\ker\psi = \langle x^2 + 1 \rangle$ F.I.T shows $\mathbb{Z}[x]/\langle x^2+1 \rangle \cong \mathbb{Z}[i]$

$x^3 - 3x^2 + 2x + 7 + \langle x^2 + 1 \rangle \overset{?}{=} a + bi$

Think of things in the ideal as $0 \Rightarrow x^2 + 1 = 0 \Rightarrow x^2 = -1$

$x(-1) - 3(-1) + 2x + 7 + \langle x^2+1 \rangle = x + 10 + \langle x^2 + 1 \rangle \xmapsto{x \to i} i + 10$

Ex. $\varphi : \mathbb{Z}[x] \longrightarrow \mathbb{R}$ by $\varphi(f(x)) = f(\sqrt[3]{2})$

$\varphi(\mathbb{Z}[x]) = \{a + b\sqrt[3]{2} + c(\sqrt[3]{2})^2 : a, b, c \in \mathbb{Z}\} = \mathbb{Z}[\sqrt[3]{2}]$, example not surjective

$\ker\varphi = \langle x^3 - 2 \rangle$, so F.I.T says $\mathbb{Z}[x]/\langle x^3 - 2 \rangle \cong \mathbb{Z}[\sqrt[3]{2}]$